

数理统计的基本概念及参数估计

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1. (1) 由简单随机样本的独立性知：

$$\begin{aligned} f_{X_1, X_2, \dots, X_n}(x_1, x_2, \dots, x_n) &= \prod_{i=1}^n f_{X_i}(x_i) \\ &= \begin{cases} \lambda^n e^{-\lambda \sum_{i=1}^n x_i}, & \forall i, x_i \geq 0 \\ 0, & \text{else} \end{cases} \end{aligned}$$

$$\mathbb{E}(\bar{X}) = \frac{1}{\lambda}, D(\bar{X}) = \frac{1}{n\lambda^2}$$

(2) 由题知： $\hat{\lambda} = \frac{n}{\sum_{i=1}^n X_i}$. 令 $X = \sum_{i=1}^n X_i, Y = \frac{1}{X}$, 则

$$X \sim \Gamma(n, \lambda)$$

由密度变换公式：

$$\begin{aligned} f_Y(y) &= f_X(x) \left| \frac{dx}{dy} \right| \\ &= \frac{1}{y^2} f_X\left(\frac{1}{y}\right) \\ &= \frac{\lambda^n y^{-n-1} e^{-\lambda y^{-1}}}{\Gamma(n)} \end{aligned}$$

故

$$\begin{aligned} \mathbb{E}(Y) &= \frac{\lambda^n}{\Gamma(n)} \int_0^{+\infty} y^{-n} e^{-\lambda y^{-1}} dy \\ &= \frac{-\lambda^n}{\Gamma(n)} \int_{+\infty}^0 t^{n-2} e^{-\lambda t} dt \quad (y = t^{-1}) \\ &= \frac{1}{\Gamma(n)} \frac{\lambda^n}{\lambda^{n-1}} \int_0^{+\infty} m^{n-2} e^{-m} dm \quad (m = \lambda t) \\ &= \frac{\Gamma(n-1)}{\Gamma(n)} \lambda \\ &= \frac{\lambda}{n-1} \end{aligned}$$

$$\text{同理, } \mathbb{E}(Y^2) = \frac{\lambda^n}{\Gamma(n)} \int_0^{+\infty} y^{-n+1} e^{-\lambda y^{-1}} dy = \frac{\lambda^2}{(n-1)(n-2)}.$$

$$\text{因此 } D(Y) = \mathbb{E}(Y^2) - \mathbb{E}^2(Y) = \frac{\lambda^2}{(n-1)^2(n-2)}.$$

综上, $\mathbb{E}(\hat{\lambda}) = n\mathbb{E}(Y) = \frac{n}{n-1}\lambda, D(\hat{\lambda}) = n^2 D(Y) = \frac{n^2 \lambda^2}{(n-1)^2(n-2)}, n > 2$. 该统计量对 λ 是有偏的.

2. (1) 注意到

$$\frac{X_1 + X_2 + X_3}{\sqrt{3}} \sim N(0, 1), \frac{X_4 + X_5 + X_6}{\sqrt{3}} \sim N(0, 1).$$

因此, 当 $C = \frac{1}{3}$ 时, $CY \sim \chi^2(2)$.

(2) 注意到

$$\frac{X_1 + X_2}{\sqrt{2}} \sim N(0, 1), X_3^2 + X_4^2 + X_5^2 \sim \chi^2(3)$$

因此, 当 $C = \frac{1/\sqrt{2}}{1/\sqrt{3}} = \frac{\sqrt{6}}{2}$ 时, $Y \sim t(3)$.

(3) 由 t 分布的定义知: 存在 $Z \sim N(0, 1), Y \sim \chi^2(n)$ 满足 $X = \frac{Z}{\sqrt{\frac{Y}{n}}}$.

故 $X^2 = \frac{Z^2}{\frac{Y}{n}} \sim F(1, n)$. □

3. (1) 由题可知: 最大似然函数 $L(p) = p^n, 0 \leq p \leq 1$. 显然当 $p = 1$ 时, $L(p)$ 达到最大值. 故 p 的最大似然估计值为 1.

(2) 由题可知:

$$P(X_1 = X_2 = \cdots = X_n = 1) = \int_0^1 f(1, 1, \dots, 1, p) = \int_0^1 p^n dp = \frac{1}{n+1}$$

$$\pi(p|X_1 = X_2 = \cdots = X_n = 1) = \frac{f(1, 1, \dots, 1, p)}{P(X_1 = X_2 = \cdots = X_n = 1)} = (n+1)p^n$$

(3) 由题可知:

$$\mathbb{E}(p|X_1 = X_2 = \cdots = X_n = 1) = \int_0^1 (n+1)p^{n+1} dp = \frac{n+1}{n+2}$$

(4) 我认为贝叶斯学派对 p 的估计值更合理. 当 $n \rightarrow +\infty$ 时, 两个估计值的极限值相等.

4. (1) 一阶矩 $A_1 = \int_c^{+\infty} \theta c^\theta x^{-\theta} dx = \frac{\theta c}{\theta-1}$. 解得 $\theta = \frac{A_1}{A_1-c}$.
故矩估计量 $\hat{\theta}(X_1, X_2, \dots, X_n) = \frac{\bar{X}}{\bar{X}-c}$, 其中 $\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$.
矩估计值 $\hat{\theta}(x_1, x_2, \dots, x_n) = \frac{\bar{x}}{\bar{x}-c}$, 其中 $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$.

(2) 一阶矩 $A_1 = \int_0^1 \sqrt{\theta} x^{\sqrt{\theta}} dx = \frac{\sqrt{\theta}}{\sqrt{\theta}+1}$. 解得 $\theta = \frac{A_1^2}{(1-A_1)^2}$.
故矩估计量 $\hat{\theta}(X_1, X_2, \dots, X_n) = \frac{\bar{X}^2}{(1-\bar{X})^2}$, 其中 $\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$.
矩估计值 $\hat{\theta}(x_1, x_2, \dots, x_n) = \frac{\bar{x}^2}{(1-\bar{x})^2}$, 其中 $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$.

(3) 一阶矩 $A_1 = mp$. 解得 $p = \frac{A_1}{m}$.
故矩估计量 $\hat{p}(X_1, X_2, \dots, X_n) = \frac{\bar{X}}{m}$, 其中 $\bar{X} = \frac{\sum_{i=1}^n X_i}{n}$.
矩估计值 $\hat{p}(x_1, x_2, \dots, x_n) = \frac{\bar{x}}{m}$, 其中 $\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$.

5. (1) 由题知:

$$L(\mu, \delta) = P(X_1 = x_1, X_2 = x_2, \dots, X_n = x_n | \mu, \delta) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\delta}} e^{-\frac{(x_i - \mu)^2}{2\delta}}$$

$$\ln L(\mu, \delta) = -\frac{n}{2} \ln(2\pi\delta) - \frac{1}{2\delta} \sum_{i=1}^n (x_i - \mu)^2$$

$$\frac{\partial \ln L}{\partial \mu} = \frac{1}{\delta} \sum_{i=1}^n (x_i - \mu)$$

$$\frac{\partial \ln L}{\partial \delta} = -\frac{n}{2\delta} + \frac{1}{2\delta^2} \sum_{i=1}^n (x_i - \mu)^2$$

故当 $\mu = \frac{1}{n} \sum_{i=1}^n x_i, \delta = \frac{1}{n} \sum_{i=1}^n (x_i - \mu)^2$ 时, $L(\mu, \delta)$ 取到最大值.

故最大似然估计量 $(\hat{\mu}, \hat{\delta}) = (\frac{1}{n} \sum_{i=1}^n X_i, \frac{1}{n} \sum_{i=1}^n (X_i - \hat{\mu})^2)$. 最大似然估计值 $(\hat{\mu}, \hat{\delta}) = (\frac{1}{n} \sum_{i=1}^n x_i, \frac{1}{n} \sum_{i=1}^n (x_i - \hat{\mu})^2)$.

由 Khinchin 大数定律:

$$\hat{\mu} \xrightarrow{P} \mathbb{E}(X_i) = \mu, (n \rightarrow +\infty)$$

$$\hat{\delta} \xrightarrow{P} \mathbb{E}((X_i - \mu)^2) = \delta \mathbb{E} \left[\left(\frac{X_i - \mu}{\sqrt{\delta}} \right)^2 \right] = \delta, (n \rightarrow +\infty)$$

故 $\hat{\mu}, \hat{\delta}$ 均是相合的.

(2) 显然, 总体的一阶矩 $A_1 = \mu$, 二阶中心矩 $B_2 = \delta$.

故 $\hat{\mu} = \frac{1}{n} \sum_{i=1}^n X_i, \hat{\delta} = \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2$. 由 (1) 知它们是相合的.

(3) 根据 Fisher 信息的定义: $I(\mu) = \mathbb{E} \left[\left(\frac{\partial \ln f}{\partial \mu} \right)^2 \right] = -\mathbb{E} \left[\frac{\partial^2 \ln f}{\partial \mu^2} \right]$, 其中 f 为概率密度函数, 第二个等号成立当且仅当 $\mathbb{E} \left(\frac{\partial \ln f}{\partial \mu} \right) = 0$.

由题知:

$$\begin{aligned} f_X(x, \mu) &= \frac{1}{\sqrt{2\pi\delta}} e^{-\frac{(x-\mu)^2}{2\delta}} \\ \ln f &= -\frac{1}{2} \ln(2\pi\delta) - \frac{(x-\mu)^2}{2\delta} \\ \frac{\partial \ln f}{\partial \mu} &= \frac{1}{\delta} (x - \mu) \end{aligned}$$

故 $I(\mu) = \frac{1}{\delta}$. 另一方面, $D(\hat{\mu}) = \frac{\delta}{n} = \frac{1}{nI(\mu)}$. 故对 μ 的无偏估计达到了 Cramer-Rao 不等式的下界.

(4) 令 $\lambda = \mu^2$, $\hat{\lambda} = \hat{A}_2 - S_n$, 其中 A_2 为二阶矩, S_n 为样本方差. 注意到

$$\mathbb{E}(\hat{\lambda}) = \mathbb{E}(\hat{A}_2) - \mathbb{E}(S_n) = A_2 - \delta = \lambda$$

故 $\hat{\lambda} = \hat{A}_2 - S_n$ 是 μ^2 的无偏估计.

6. (1) 注意到 $Y = 2\lambda n \bar{X} = 2\lambda \sum_{i=1}^n X_i$, 其中 $\sum_{i=1}^n X_i \sim \Gamma(n, \lambda)$. 由 Γ 分布的性质:

$$Y \sim \Gamma(n, \frac{1}{2}) = \Gamma(\frac{2n}{2}, \frac{1}{2}) = \chi^2(2n)$$

□

由第 1 题知: $\mathbb{E}(\frac{1}{\bar{X}}) = \frac{n}{n-1} \lambda$.

(2) 由 (1) 知 $\phi(X_1, X_2, \dots, X_n) = \frac{n-1}{n\bar{X}}$ 为 λ 的一个无偏估计.

7. 设随机变量 $X_i \sim B(1, p)$, $X_i = 1 \iff$ 第 i 件货物为次品. 令 $X = \sum_{i=1}^n X_i$, 则由中心极限定理:

$$\frac{X - np}{\sqrt{np(1-p)}} \approx Z \sim N(0, 1)$$

故 $P\left(-z_{0.975} < \frac{X - np}{\sqrt{np(1-p)}} < z_{0.975}\right) = 0.95$, 解得

$$\frac{-b - c}{2a} < p < \frac{-b + c}{2a}$$

其中 $a = n(n + z_{0.975}^2) = 1003841.6$, $b = -n(2X + z_{0.975}^2) = -303841.6$,
 $c = \sqrt{n^2 z_{0.975}^4 + 4n^2 z_{0.975}^2 X - 4n z_{0.975}^2 X^2} \approx 44429.4$.

故置信水平为 0.95 的次品率 p 的置信区间为 (0.1292, 0.1735).

8. 设正教授的住房面积总体为 X_1 , 样本为 $X_{11}, X_{12}, \dots, X_{1n}$, 副教授的住房面积总体为 X_2 , 样本为 $X_{21}, X_{22}, \dots, X_{2n}$, $n = 60$. 令 $\bar{X}_k = \frac{1}{n} \sum_{i=1}^n X_{ki}$, $S_k^2 = \frac{1}{n-1} \sum_{i=1}^n (X_{ki} - \bar{X}_k)$, $k = 1, 2$.

(1) 由于总体方差未知, 故选取 $\frac{\sqrt{n}(\bar{X}_1 - \mu_1)}{S_1} \sim t(n-1)$ 作为枢轴量, 其中 $\mu_1 = \mathbb{E}(X_1)$. 故

$$P(-t_{0.975} < \frac{\sqrt{n}(\bar{X}_1 - \mu_1)}{S_1} < t_{0.975}) = 0.95$$

解得置信水平为 0.95 的 μ_1 的置信区间为

$$\left(\bar{X}_1 - \frac{t_{0.975} S_1}{\sqrt{n}}, \bar{X}_1 + \frac{t_{0.975} S_1}{\sqrt{n}}\right)$$

其中 $\bar{X}_1 = 115.4$, $S_1 = 15.8$, $t_{0.975}$ 为常数.

(2) 与 (1) 同理可得置信区间为

$$\left(\bar{X}_2 - \frac{t_{0.975} S_2}{\sqrt{n}}, \bar{X}_2 + \frac{t_{0.975} S_2}{\sqrt{n}}\right)$$

其中 $\bar{X}_2 = 89.3$, $S_2 = 21.3$, $t_{0.975}$ 为常数.

(3) 由于总体方差已知, 可选取 $\frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2 + \sigma_2^2}{n}}} \sim N(0, 1)$ 作为枢轴量. 其中 μ_k, σ_k^2 分别为总体 X_k 的均值和方差, $k = 1, 2$. 故

$$P(-z_{0.975} < \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2 + \sigma_2^2}{n}}} < z_{0.975}) = 0.95$$

解得置信水平为 0.95 的 $\mu_1 - \mu_2$ 的置信区间为

$$\left((\bar{X}_1 - \bar{X}_2) - z_{0.975} \sqrt{\frac{\sigma_1^2 + \sigma_2^2}{n}}, (\bar{X}_1 - \bar{X}_2) + z_{0.975} \sqrt{\frac{\sigma_1^2 + \sigma_2^2}{n}}\right)$$

其中 $\bar{X}_1, \bar{X}_2$ 同上, $\sigma_1 = 15.8, \sigma_2 = 21.3$, $z_{0.975}$ 为常数.

9. (1) 根据期望和方差的定义:

$$\begin{aligned}
 \mathbb{E}_\theta[z(X, \theta)] &= \mathbb{E}[z(X|\theta)] \\
 &= \int_{-\infty}^{+\infty} \left(\frac{\partial}{\partial \theta} \ln f(x|\theta) \right) f(x|\theta) dx \\
 &= \int_{-\infty}^{+\infty} \left(\frac{1}{f(x|\theta)} \frac{\partial}{\partial \theta} f(x|\theta) \right) f(x|\theta) dx \\
 &= \frac{\partial}{\partial \theta} \int_{-\infty}^{+\infty} f(x|\theta) dx \\
 &= 0
 \end{aligned}$$

$$\begin{aligned}
 D_\theta[z(X, \theta)] &= \mathbb{E}[z^2(X|\theta)] \\
 &= \int_{-\infty}^{+\infty} \left(\frac{\partial}{\partial \theta} \ln f(x|\theta) \right)^2 f(x|\theta) dx \\
 &= \int_{-\infty}^{+\infty} \frac{1}{f(X|\theta)} \left(\frac{\partial f(x|\theta)}{\partial \theta} \right)^2 dx
 \end{aligned}$$

另一方面,

$$\begin{aligned}
 \mathbb{E}_\theta \left[\frac{\partial z(X, \theta)}{\partial \theta} \right] &= \mathbb{E} \left[\frac{\partial z(X|\theta)}{\partial \theta} \right] \\
 &= \int_{-\infty}^{+\infty} \left(\frac{\partial^2}{\partial \theta^2} \ln f(X|\theta) \right) f(X|\theta) dx \\
 &= \int_{-\infty}^{+\infty} \frac{\partial}{\partial \theta} \left(\frac{1}{f(X|\theta)} \frac{\partial f(X|\theta)}{\partial \theta} \right) f(X|\theta) dx \\
 &= \int_{-\infty}^{+\infty} -\frac{1}{f(X|\theta)} \left(\frac{\partial f(X|\theta)}{\partial \theta} \right)^2 dx + \int_{-\infty}^{+\infty} \frac{\partial^2 f(X|\theta)}{\partial \theta^2} dx \\
 &= -D_\theta[z(X, \theta)] + \frac{\partial^2}{\partial \theta^2} \int_{-\infty}^{+\infty} f(X|\theta) dx \\
 &= -D_\theta[z(X, \theta)]
 \end{aligned}$$

□

(2) 由题可知: $\hat{\theta} \xrightarrow{P} \theta_0, (n \rightarrow +\infty)$. 故可将 $l'(\theta)$ 在 $\hat{\theta}$ 处展开:

$$l'(\hat{\theta}) \approx l'(\theta_0) + l''(\theta_0)(\hat{\theta} - \theta_0)$$

由于 $\hat{\theta}$ 是参数 θ_0 的最大似然估计, 故 $l'(\hat{\theta}) = 0$, 因此

$$\hat{\theta} - \theta_0 \approx -\frac{l'(\theta_0)}{l''(\theta_0)}$$

也即

$$\sqrt{n}(\hat{\theta} - \theta_0) \approx -\frac{l'(\theta_0)/\sqrt{n}}{l''(\theta_0)/n}$$

□

(3) 注意到

$$\frac{l'(\theta_0)}{\sqrt{n}} = \frac{1}{\sqrt{n}} \sum_{i=1}^n \frac{\partial f(X|\theta)}{\partial \theta} \Big|_{\theta=\theta_0} = \frac{1}{\sqrt{n}} \sum_{i=1}^n z(X_i|\theta_0)$$

由 (1) 以及中心极限定理可知：当 $n \rightarrow +\infty$ 时，

$$\frac{\frac{\sum_{i=1}^n z(X_i|\theta) - 0}{n}}{\sqrt{\frac{\mathbb{E}\left[\frac{\partial z(X|\theta)}{\partial \theta}\right]}{n}}} \xrightarrow{d} Z \sim N(0, 1)$$

故

$$\frac{l'(\theta_0)}{\sqrt{n}} \xrightarrow{d} \sqrt{-\mathbb{E}\left[\frac{\partial z(X|\theta)}{\partial \theta}\right]_{\theta=\theta_0}} Z \sim N(0, -\mathbb{E}\left[\frac{\partial z(X|\theta)}{\partial \theta}\right]_{\theta=\theta_0})$$

另一方面，由大数定律：当 $n \rightarrow +\infty$ 时，

$$\frac{1}{n} l''(\theta_0) = \frac{1}{n} \sum_{i=1}^n \frac{\partial z(X|\theta)}{\partial \theta} \bigg|_{\theta=\theta_0} \xrightarrow{P} \mathbb{E}\left[\frac{\partial z(X|\theta)}{\partial \theta} \bigg|_{\theta=\theta_0}\right]$$

故由 (2) 以及 Slutsky 引理：当 $n \rightarrow +\infty$ 时，

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N\left(0, -\frac{1}{\mathbb{E}\left[\frac{\partial z(X|\theta)}{\partial \theta} \bigg|_{\theta=\theta_0}\right]}\right) = N\left(0, \frac{1}{I(\theta_0)}\right)$$

□

10. (1) 根据分布函数的单调性可知：对 $\forall x \in [x_{M,k}, x_{M,k+1})$,

$$F_n(X_{M,k+1}; \omega) - F_n(x; \omega) + \frac{1}{M} \geq \frac{1}{M} = F(X_{M,k+1}) - F(X_{M,k}) \geq F(X_{M,k+1}) - F(x)$$

故第一个不等式得证. 同理可得对 $\forall x \in [x_{M,k}, x_{M,k+1})$,

$$F_n(x; \omega) - F_n(X_{M,k}; \omega) + \frac{1}{M} \geq \frac{1}{M} = F(X_{M,k+1}) - F(X_{M,k}) \geq F(x) - F(X_{M,k})$$

□

(2) 由题可知：

$$\begin{aligned} P\left(\sup_{x \in \mathbb{R}} |F_n(x; \omega) - F(x)| > \frac{2}{M}\right) &\leq P\left(\max_{1 \leq k \leq M} |F_n(x; \omega) - F(x)| + \frac{1}{M} > \frac{2}{M}\right) \\ &= P\left(\max_{1 \leq k \leq M} |F_n(x; \omega) - F(x)| > \frac{1}{M}\right) \\ &\leq P\left(\sum_{k=1}^M |F_n(x; \omega) - F(x)| > \frac{1}{M}\right) \\ &\leq \sum_{k=1}^M P\left(|F_n(x; \omega) - F(x)| > \frac{1}{M^2}\right) \\ &\xrightarrow{P} 0, (n \rightarrow +\infty) \end{aligned}$$

□